CHERI ACKERMAN (NÉE FAVEL)

Chemical / Process Engineer (MEng) | Industrial R&D · Hydrometallurgy · Water Treatment

Abu Dhabi, UAE · +971 58 584 5473 · cherifavel@gmail.com · linkedin.com/in/cheri-favel

PROFILE

MEng Chemical Engineer with industry and postgraduate experience in process development, hydrometallurgy, electrochemical systems, mineral processing and industrial water treatment. Skilled in experimental planning, process calculations, mass and energy balances, PFD/P&ID preparation, technical reporting and analytical data interpretation. Contributed to client-focused engineering investigations and published peer-reviewed research.

CORE TECHNICAL STRENGTHS

- **Process engineering:** process design support, mass/energy balances, process calculations, equipment-sizing support, PFDs/P&IDs, design bases and specifications, CAPEX/OPEX support.
- **Hydrometallurgy / electrochemistry:** mineral leaching, electrowinning, metal recovery, selective precipitation, manganese processing and reactor testing.
- **Water / environmental:** industrial tailings-water treatment, Fe/Zn/Mn removal, solid–liquid separation, water chemistry; familiarity with UF/NF/RO fundamentals.
- **Laboratory / data:** ICP-OES, XRF, XRD, FTIR, SEM-EDS, BET, TGA; experimental design, kinetic modelling and data interpretation.
- **Software:** advanced Excel (VBA, Power Query, PivotTables), Visio, Python; Aspen Plus/HYSYS foundational exposure.

ENGINEERING EXPERIENCE

Process & R&D Engineer — Free Radical Process Design (FRPD), South Africa **Mar 2021 – Jul 2022**

- Conducted experimental and engineering investigations for hydrometallurgical, electrochemical and water-treatment process development and optimisation.
- Performed mass/energy balances and process calculations; supported unit design and equipment sizing; prepared PFDs/P&IDs, design bases, specifications and reports.
- Planned and coordinated client laboratory work, analysed test results and developed technically supported recommendations; reviewed supplier quotations.

SELECTED ENGINEERING & R&D PROJECTS

- **Tailings-water treatment, Peru:** Investigated Fe, Zn and Mn removal/recovery by oxidation, precipitation and electrolysis; evaluated treatment performance and feasibility.
- **Continuous cobalt electrowinning:** Supported development and technical evaluation of a continuous electrochemical recovery process.
- **Manganese processing:** Supported manganese sulphate monohydrate process development and investigated selective precipitation and electrochemical beneficiation routes.
- **Gold recovery reactor:** Contributed to lab-scale reactor development for residual gold recovery from refinery eluate; investigated voltage, current density, flow and reaction time.
- **Rotating-cathode zinc removal:** Investigated an electrochemical zinc-removal system for tailings water, including considerations for scaling toward mobile-plant application.
- **Phlogopite leaching / potassium recovery (MEng):** Designed nitric-acid leaching experiments, modelled kinetics and characterised mineral feed and residues; findings published in Clays and Clay Minerals.
- **Additional investigations:** Fe/Al precipitation from leach liquor; Ni-MH battery metal-recovery feasibility; neodymium plant CAPEX/OPEX analysis; galvanising-process review.

ADDITIONAL PROFESSIONAL EXPERIENCE

Business Setup & Marketing Consultant — Cresco Consulting FZCO, UAE **Apr 2025 – Present**

- Prepare and coordinate business setup documentation, maintain records and support client communications and marketing activities.

Vessel Operations & Engineering Support — Research Vessel Rosa, USA & Bahamas **May 2023 – Apr 2025**

- Assisted with vessel equipment checks and troubleshooting involving pumps, plumbing and RO watermakers; followed marine safety procedures.

Academic Journal Editor — CACTUS Communications (Remote) **Nov 2022 – Apr 2023**

- Edited scientific manuscripts for technical clarity, consistency and publisher formatting requirements.

Postgraduate Researcher — University of Pretoria, South Africa **Jan 2020 – Dec 2021**

- Independently planned and conducted two years of laboratory investigations, including experimental methodology, kinetic modelling, analytical characterisation, data interpretation and research reporting.
- Maintained experimental records, supervised research assistants and communicated results through reports and peer-reviewed publication.

Teaching Assistant / Laboratory Intern — University of Pretoria **2017 – 2021**

- Supported engineering laboratory instruction and completed mineral-leaching and fertiliser-related laboratory investigations.

EDUCATION & PUBLICATION

- MEng Chemical Engineering — University of Pretoria (2021); published dissertation research in Clays and Clay Minerals.
- BEngHons Chemical Engineering, Distinction — University of Pretoria (2020); Academic Honorary Colours.
- BEng Chemical Engineering — University of Pretoria (2019); Environmental Engineering specialisation.

SELECTED TRAINING & CERTIFICATIONS

- LEED Green Associate Exam Prep — LinkedIn Learning (2026); Project Management: Energy & Water Projects — PMI / LinkedIn Learning (2026).
- Occupational Safety & Health: Hazard Communication — LinkedIn Learning (2026); STCW Basic Safety Training (2023); Approved Engine Course AEC 1 (2024).

ADDITIONAL INFORMATION

South African · UAE resident visa · Available immediately · Languages: English (native), Afrikaans (proficient), Hebrew (basic)